

Shiva Tarun Nangunoori

Data Engineer

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Summary

Experienced Data Engineer with 3+ years of strong background in designing and optimizing large-scale data pipelines, real-time processing, and cloud-based analytics. Skilled in big data technologies like Apache Kafka, Flink, Spark, and Ignite to enable low-latency data solutions. Proficient in Python (PyArrow, Dask, Modin), SQL (Recursive CTEs, Lateral Joins), and cloud platforms (AWS, Azure). Expertise in ETL workflows, data modeling, fraud detection, and risk assessment, driving data-driven decision-making across financial, insurance, and retail domains.

Technical Skills

- Big Data & Streaming:** Apache Kafka, Flink, Spark, Hive, Trino, Hadoop, Apache Ignite, Apache Pulsar
- Programming & Data Processing:** Python (PyArrow, Dask, Modin, Pandas, PySpark), SQL (Recursive CTEs, Lateral Joins, Window Functions, Dynamic SQL), Java, Scala
- Cloud & Data Warehousing:** AWS (Kinesis, Glue, Redshift, S3, EMR), Azure (Synapse, Data Factory, Azure SQL), Google Cloud (BigQuery, Pub/Sub)
- Databases & Storage:** PostgreSQL, Apache Druid, NoSQL (Cassandra, MongoDB), Delta Lake, HDFS
- Data Engineering & Optimization:** ETL Pipelines, Real-Time Data Processing, Fraud Detection, Risk Scoring, Financial Forecasting, Data Validation, Anomaly Detection, Data Lake Design, Data Modeling, Batch & Stream Processing
- Performance & Monitoring:** Query Optimization, Sub-10ms Latency Processing, Scalable Data Architectures, Data Lineage, Monitoring Tools (Prometheus, Grafana), Distributed Systems Performance Tuning
- DevOps & Automation:** Docker, Kubernetes, CI/CD pipelines, Infrastructure as Code (Terraform, CloudFormation)
- Security & Governance:** Data Encryption, Access Control, GDPR Compliance, Data Auditing

Professional Experience

Data Engineer, Liberty Mutual Insurance

08/2023 – Present | Remote, USA

- Developed a Real-Time Digital Twin using Apache Kafka & Flink, processing 5TB of policy, claims, and financial data daily. Reduced data ingestion latency by 60%, enhancing real-time decision-making for underwriting and risk assessments.
- Optimized large-scale data processing with Apache Ignite, achieving sub-10ms query response times for 500M+ daily transactions. Integrated PostgreSQL, ensuring 99.99% uptime for claims, policy updates, and compliance reports.
- Enhanced data transformations using Python (PyArrow, Dask, Modin), improving computational efficiency by 3x. Enabled instantaneous analytics on petabyte-scale datasets, accelerating fraud detection, claims assessment, and regulatory compliance.
- Built advanced SQL models with Recursive CTEs & Lateral Joins, reducing query execution time by 45%. Delivered real-time insights for actuarial teams, optimizing premium pricing models and financial risk analysis.
- Leveraged AWS Kinesis, Glue & Redshift, reducing infrastructure costs by 30% while maintaining sub-second query performance. Improved claims processing speed, ensuring faster payouts and accurate financial forecasting.
- Designed a monitoring system with Apache Druid, Trino & Delta Lake, detecting 1B+ daily anomalies. Enhanced fraud detection by 50%, strengthening compliance and mitigating financial risks for the organization.

Data Engineer, NTT DATA Services

01/2021 – 08/2022 | Hyderabad, India

- Developed a Retail Sales Data Pipeline, processing 2TB+ of sales data daily using Apache Spark and Hadoop, ensuring 99.9% data accuracy and enabling real-time insights for inventory optimization and sales trend forecasting.
- Collaborated with cross-functional teams, including business analysts, data scientists, and stakeholders, to gather technical needs, optimize ETL workflows, and design data models, improving data accessibility and reducing reporting latency by 40%.
- Built real-time data pipelines using Apache Kafka, streaming millions of transactions with sub-100ms latency for fraud detection, personalized customer references, and dynamic pricing for enhanced operational efficiency in large-scale retail firm.
- Optimized batch processing workflows with Apache Hive and Hadoop, implementing partitioning, bucketing, and indexing, reducing query execution time by 50% and improving accessibility for financial reporting and operational analytics teams.
- Designed and optimized SQL queries using CTEs, Window Functions, and Dynamic SQL, improving data retrieval performance by 45%, reducing delay reporting, and ensuring accurate revenue forecasting and trend analysis for decision-makers.
- Developed and automated data validation scripts using Python (Pandas, PySpark), ensuring data integrity across structured and semi-structured datasets, reducing data inconsistencies by 35% and improving overall ETL pipeline reliability.
- Integrated structured and semi-structured data using Azure Data Factory and Azure Synapse Analytics, reducing ETL processing time by 55%, ensuring seamless BI integration, and enabling real-time insights for executives and business teams.

Education

Master of Science, University of Maryland Baltimore County
Information Systems

09/2022 - 05/2024 | Maryland, USA

Projects

Project 1: Intelligent Insurance Risk Scoring System

- Engineered a risk-scoring engine using Apache Kafka & Flink, analyzing 4TB+ of claims and policy data daily to generate real-time risk scores, improving fraud detection accuracy by 55% and optimizing underwriting decisions.
- Integrated a predictive alert system with Apache Druid & Trino, flagging high-risk claims within 10ms, reducing fraudulent payouts by 40%, and ensuring faster claim resolution while maintaining compliance with insurance regulations.

Project 2: AI-Driven Retail Demand Forecasting

- Developed an advanced demand forecasting pipeline using Apache Spark & Delta Lake, processing 1.5TB+ of retail sales data daily, reducing stock wastage by 35% and improving inventory turnover rates for nationwide retail chains.
- Optimized ETL workflows with Azure Synapse & Data Factory, automating real-time data ingestion from multiple sources, cutting reporting latency by 60%, and enabling dynamic pricing adjustments based on real-time sales trends.